



Operations Research

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Exercise Sheet 6

Training Exercises

Exercise T6.1. *Knapsack Branch-and-Bound*

We consider the knapsack problem with $n = 4$ items $i \in \{1, \dots, n\}$, where item i has value $v_i \in \mathbb{N}$ and weight $w_i \in \mathbb{N}$. The backpack has total capacity of $c = 5$. The problem can be formulated as an IP as follows:

$$\begin{aligned} \max \quad & \sum_{i=1}^n v_i x_i \\ \text{s.t.} \quad & \sum_{i=1}^n w_i x_i \leq c \\ & x_i \in \{0, 1\} \forall i = 1, \dots, n \end{aligned}$$

where $x_i = 1$ means that item i is packed into the backpack. An LP relaxation is given by

$$\begin{aligned} \max \quad & \sum_{i=1}^n v_i x_i \\ \text{s.t.} \quad & \sum_{i=1}^n w_i x_i \leq c \\ & x_i \leq 1 \forall i = 1, \dots, n \\ & x_i \geq 0 \forall i = 1, \dots, n. \end{aligned}$$

Consider the following values and weights of the objects:

i	1	2	3	4
value $\mathbf{v_i}$	20	12	2	9
weight $\mathbf{w_i}$	2	2	1	2

Solve the knapsack problem by using the branch-and-bound method. Specifically, for each branching step, give an optimal solution, as well as the optimal objective value.

- a) Determine an optimal solution to the LP relaxation of the problem.

Hint: The LP relaxation of the general knapsack problem has the following optimal solution, which you may use without justification: Determine the value-weight ratios

$q_i = v_i/w_i$ and index the items such that $q_j \geq q_{j+1}$ for all $j = 1, \dots, n-1$. Choose k such that $\sum_{j=1}^k w_j \leq c$, but $\sum_{j=1}^{k+1} w_j > c$. Set $x_j = 1$ for $j = 1, \dots, k$, $x_{k+1} = (c - \sum_{j=1}^k w_j)/w_{k+1}$, and $x_j = 0$ for $j > k+1$.

- b) Apply the branch-and-bound algorithm to the problem to obtain an optimal solution of the IP. In each step, state the optimal solution of the respective subproblem, as well as the optimal objective value of the intermediate solutions.

Exercise T6.2. Gomory cuts

For each of the following optimal tableaus of maximization problems, identify all Gomory cuts that can be derived in the first iteration.

a)

Base	Coefficients				Res.
	x_1	x_2	s_1	s_2	
z	0	0	2	1	17/3
x_1	1	0	3/2	-1/2	5/3
x_2	0	1	-2/3	4/3	4/3

b)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	0	0	1	2	3	31/4
x_1	1	0	-1/2	2/3	0	5/4
x_2	0	1	3/2	-1/5	0	7/4
x_5	0	0	2/3	1	1	3

Exercise T6.3. Staff Scheduling

The jam factory “Waldgeist” has received several large orders for the next month and therefore need to hire additional temporary workers (temps) to meet the increased demand.

Waldgeist operates in three shifts daily (early, late and night) and there is a need for d_{st} temps on shift $s \in \{\text{early, late, night}\}$ on day $t \in \{1, \dots, 30\}$.

In response to a job posting, 20 persons $I := \{1, \dots, 20\}$ applied, all of whom are eligible.

Model an IP that you use to decide which of the temps will be employed and which will show the assignment of temps to shifts and days. After each temp has a fixed cost for the company, you want to employ as few of them as possible, but at the same time cover the demand.

However, due to legal constraints, staff scheduling must also adhere to the following rules:

- A temp must work a minimum of 5 shifts and a maximum of 15 shifts.
- A temp may work at most one shift per day.
- If a temp works the night shift, he/she may not work the morning shift on the next day.
- If a temp has a day off, they will have at least two days off in a row.
- A temp may work no more than 7 days in a row.

Hint: For the last 3 rules, it may be assumed that the temps have at least two days off both before Day 1 and after Day 30.

Self-study Exercises

Exercise S6.1. *Branch-and-Bound*

Given the following model

$$\begin{aligned} \max \quad & 17x_1 + 12x_2 \\ \text{s.t.} \quad & 10x_1 + 7x_2 \leq 40 \\ & x_1 + x_2 \leq 5 \\ & x_1, x_2 \in \mathbb{N}_0. \end{aligned}$$

Determine an optimal solution using the branch-and-bound method. (Note: To solve the (sub)problems, feel free to use Gurobi/the Internet).

Exercise S6.2. *Cutting Planes*

Given the following IP, you want to solve it using the cutting plane method.

$$\begin{aligned} \max \quad & 2x_1 + 3x_2 \\ \text{s.t.} \quad & 4x_1 + 2x_2 \leq 3 \\ & x_2 \leq 3 \\ & x_1, x_2 \in \mathbb{N}_0 \end{aligned}$$

Determine all possible Gomory cuts in the first and second iteration of the procedure. For the second iteration, use the integer form of the Gomory cut corresponding to the basic variable x_2 .